

From Endogenous Credit to Borrower Composition

A Reproducible Measure from Japanese Sectoral Loan Stocks

Abstract

Bank lending can grow while its borrower mix changes. Endogenous-credit theory distinguishes credit creation from credit allocation, but public data often show only total lending or borrower-sector stocks. We construct a reproducible borrower-composition measure from Bank of Japan sectoral loan stocks. The published Japan crosswalk of [Bezemer et al. \(2020\)](#) divides one loan population into non-financial business, financial business, property and mortgages, and household non-housing credit. Stock changes are summed within each bucket before positive parts are taken, so finer classifications within a bucket do not alter the four-quarter vector. We also report alternative placements of construction based on [Werner \(1997\)](#) and [Müller & Verner \(2024\)](#). Accounting and coverage checks support the bridge; external data provide only partial checks of selected branches. In the current-vintage pseudo-out-of-sample application, five of six main estimates have higher point loss than the matched credit-stock baseline. The remaining main estimate and one Appendix estimate are lower, but both intervals include zero. The result is a reproducible measure of borrower composition, not loan purpose or a forecasting result.

Keywords: endogenous credit; borrower composition; sectoral lending; measurement; Bank of Japan

JEL classification: E44; E51; E58; G21; G28

1 Introduction

Bank lending creates deposits and cannot be reduced to a prior stock of loanable funds ([Le Bourva, 1992](#); [Fontana, 2003](#); [McLeay et al., 2014](#)). Endogenous-money and monetary-circuit theories also make the location of credit important. Loans to non-financial firms, construction companies, financial intermediaries, and households enter different balance-sheet positions even when total lending is unchanged ([Rochon & Rossi, 2007](#); [Passarella, 2014](#); [Werner, 2012](#)). Credit scale alone does not capture this difference.

Empirical studies find different macro-financial patterns for mortgage and business credit ([Bezemer et al., 2020](#); [Bezemer & Zhang, 2019](#)), tradable and non-tradable sector credit ([Müller & Verner, 2024](#); [Andries et al., 2025](#)), and riskier corporate credit allocation ([Brandão-Marques et al., 2022](#)). Research on Japan also shows shifts and dispersion in sectoral loan growth ([Saita & Sekine, 2001](#)). Yet a basic measurement problem remains. The Bank of Japan (BOJ) reports detailed loan stocks by borrower sector, but not comprehensive gross originations classified jointly by borrower and purpose. Stock changes can reflect new lending, repayment, write-offs, or reclassification. Borrower labels therefore do not reveal what a loan financed.

This paper addresses the observable part of that problem. It constructs and audits a borrower-composition measure based on the published Japan crosswalk of [Bezemer et al. \(2020\)](#). The measure divides one included loan population into non-financial business (NFB), financial business, property and mortgages, and household non-housing credit. Credit scale is calculated from the same

population. The scalar q_t is the NFB coordinate of this four-part vector. It is not a share of credit reaching a final destination.

Construction shows why the classification must remain visible. It belongs to NFB in the primary Japan crosswalk, enters the financial-circulation proxy in [Werner \(1997\)](#), and is non-tradable in [Müller & Verner \(2024\)](#). We apply all three classifications to the same loan population and report them together. This turns a hidden classification choice into a direct robustness check. The earlier author-defined G/B/E grouping remains only in the Appendix for archive compatibility.

The BOJ mapping, coverage, residual accounting, and positive and net changes form the main evidence. MOF and MLIT data provide partial checks for manufacturing and household housing. A pseudo-out-of-sample comparison shows how the measure can be used with a matched credit-stock baseline, although it does not establish a forecast improvement in the present sample. The contribution is an auditable measurement bridge.

2 Conceptual Motivation and Measurement Design

2.1 From endogenous credit to two observable coordinates

Classical quantity theories relate nominal expenditure to money and its velocity ([Fisher, 1911](#); [Friedman, 1956](#); [Laidler, 1985](#)). Endogenous-credit accounts instead start from bank lending and deposit creation ([Le Bourva, 1992](#); [Fontana, 2003](#); [McLeay et al., 2014](#)). The Quantity Theory of Credit further emphasises that the economic use of credit conditions its transmission ([Werner, 2012](#)). Monetary-circuit theory supplies an institutional account: bank credit opens and finances circuits whose balance-sheet positions need not be equivalent ([Rochon & Rossi, 2007](#); [Passarella, 2014](#)).

The available BOJ data do not identify those uses. They do identify borrower sectors and two household-purpose components. Let B_t^{TOT} be the BOJ total for domestically licensed banks, where “domestically licensed” describes the lenders rather than borrower geography. Let B_t^{FIN} , B_t^{RE} , B_t^{LG} , B_t^{HH} , and B_t^{HOU} denote, respectively, finance and insurance, real estate, local-government, total household, and household housing stocks. The primary Japan crosswalk defines

$$L_t^{NFB} = B_t^{TOT} - B_t^{FIN} - B_t^{RE} - B_t^{LG} - B_t^{HH}, \quad (1)$$

$$L_t^{FIN} = B_t^{FIN}, \quad (2)$$

$$L_t^{PROP} = B_t^{RE} + B_t^{HOU}, \quad (3)$$

$$L_t^{HHN} = B_t^{HH} - B_t^{HOU}. \quad (4)$$

The included stock is therefore

$$L_t^P = \sum_{k \in \mathcal{K}} L_t^k = B_t^{TOT} - B_t^{LG}, \quad \mathcal{K} = \{NFB, FIN, PROP, HHN\}. \quad (5)$$

The published NFB residual contains the separately disclosed overseas-linked series and a small unresolved remainder. We retain both explicitly in the coverage audit rather than relabelling them as domestic borrower observations. The scale coordinate used at horizon h is

$$g_{t,h}^P = \Delta_h \log L_t^P. \quad (6)$$

Thus scale and composition refer to the same lenders, loan instrument, and borrower population.

For composition, first form each bucket’s net change and positive part,

$$\Delta L_t^k = L_t^k - L_{t-1}^k, \quad P_t^k = \max\{\Delta L_t^k, 0\}, \quad C_t = \sum_{\ell \in \mathcal{K}} P_t^\ell. \quad (7)$$

For a window of w quarters, define

$$P_{t,w}^k = \sum_{j=0}^{w-1} P_{t-j}^k, \quad \pi_{t,w}^k = \frac{P_{t,w}^k}{\sum_{\ell \in \mathcal{K}} P_{t,w}^\ell}. \quad (8)$$

The primary empirical object is the four-quarter vector $\pi_{t,4} = (\pi_{t,4}^{NFB}, \pi_{t,4}^{FIN}, \pi_{t,4}^{PROP}, \pi_{t,4}^{HHN})$. The scalar $q_t \equiv \pi_{t,4}^{NFB}$ is a display coordinate. Four-quarter aggregation reduces quarter-specific stock-difference noise and remains defined whenever the four-quarter positive-flow denominator is positive. Neither the vector nor q_t is a share of credit used for production, investment, or asset purchases.

2.2 Granularity invariance and claim boundary

Suppose a component of L_t^k is divided into additive subseries $s_1, \dots, s_m$ that remain in bucket k . Then

$$\sum_{j=1}^m \Delta L_{s_j,t} = \Delta L_{s,t}.$$

Because the positive part in equation (7) is applied after this sum, P_t^k , $\pi_{t,w}^k$, and q_t are unchanged. The measure is therefore invariant to classification refinements within a fixed borrower bucket. It is not invariant to moving a sector across buckets. That remaining classification uncertainty is addressed by three literature-anchored mappings reported jointly rather than by presenting a single author choice as uniquely implied by the data.

This construction differs from $\sum_{s \in \mathcal{S}_k} \max\{\Delta L_{s,t}, 0\}$. The latter rises when offsetting expansions and contractions are published as separate subseries and therefore depends on classification granularity. The empirical audit reports the difference between the two orders rather than silently treating them as equivalent.

Table 1 links each observable to the check needed for its interpretation. A purpose interpretation would require independent data that code loan purpose, which the current BOJ comparator cannot provide.

Table 1: Concept-to-measure mapping and claim-boundary checks.

Object	Observable	Required check	Supported interpretation
Credit scale	$g_{t,h}^P = \Delta_h \log L_t^P$	exact same-population stock identity	growth of included BOJ loan stocks
Borrower composition	$\pi_{t,4}$, with $q_t = \pi_{t,4}^{NFB}$	four buckets, sum-then-positive	composition of positive bucket-level stock changes
Net reallocation	$(\Delta L_t^k)_{k \in \mathcal{K}}$	retain expansions and contractions	signed changes, not bounded shares
Taxonomy robustness	three literature-anchored placements	same population; jointly reported	sensitivity to cross-bucket assignment
Household residual	BOJ consumer-purpose stock	same-provider level comparison	limited check, not external validation
Outcome use	expanding-window loss comparison	matched-scale baseline and uncertainty interval	application, not construct validation

3 Data and Empirical Design

3.1 BOJ source, population, and timing

The source is the BOJ Time-Series Data Search, database LA01. The selected map is the total outstanding loans of domestically licensed banks less local-government loans. The lender designation

does not restrict borrowers to Japan: the published NFB residual includes the disclosed overseas-linked series. BOJ documentation states that the sector data have a discontinuity in 2009Q2 following an industrial-classification revision ([Bank of Japan, n.d.](#)). A change ending in 2009Q2 crosses that break, so the first valid within-taxonomy change is 2009Q3. The first complete four-quarter measure is dated 2010Q2 and becomes usable in the application in 2010Q3 under an assumed fixed 90-day release lag.

The long-term JGB yield, nominal GDP, and BOJ total assets are proxy outcomes used only to demonstrate the bridge; none measures borrower composition or validates it. Table 5 reports the JGB application, nominal GDP is not tabulated, and BOJ balance-sheet acceleration is confined to the Appendix.

3.2 Mapping and coverage

Table 2: Literature-anchored BOJ credit-allocation crosswalk.

BOJ input / component	Bridge role	Construction	Main measurement risk	Primary / explicit flow share
Published residual NFB	Japan P_t^{NFB}	Official total less finance/insurance, real estate, local government, and total household loans; construction is included.	The residual also absorbs the separately disclosed overseas-linked and unresolved components.	40.1%
Finance and insurance	plus P_t^{FIN}	Direct BOJ borrower-sector match to financial business.	Borrower sector does not identify the use of funds.	17.9%
Real estate household housing	plus P_t^{PROP}	Published Japan property/mortgage crosswalk.	Combines a borrower-sector stock with a household-purpose stock.	40.0%
Household total household housing	less P_t^{HHN}	Published residual construction for household non-housing credit.	Includes tax and other household borrowing; it is not labelled consumer credit.	2.1%
Local governments	excluded, explicit	Removed by the published Japan residual formula.	Exclusion changes the population relative to the official total.	6.0%
Overseas-linked loans	NFB component, explicit	Disclosed because the current BOJ total makes it part of residual NFB.	Not a domestic borrower-sector observation.	5.9%
Unresolved residual	NFB component, explicit	Residual after mapped industries and overseas-linked loans.	Has no independent borrower or purpose label.	0.0%

The primary mapping follows Bezemer, Samarina, and Zhang (2020); $\mathbf{q}_t$ is its four-quarter NFB coordinate. Construction remains in NFB, and residual components are shown separately. For the four primary buckets, the displayed share is $\sum_t P_t^k / \sum_t (P_t^{NFB} + P_t^{FIN} + P_t^{PROP} + P_t^{HHN})$ over the 67 valid within-taxonomy quarterly changes (2009Q3–2026Q1). Explicit component rows use the denominator stated by their construction. Positive parts are taken after net changes are aggregated within each fixed bucket.

Table 2 defines the four buckets and their common population, $L_t^P = B_t^{TOT} - B_t^{LG}$. Accounting coverage reconciles this population and the local-government exclusion to the official aggregate. Decomposition coverage accounts for the detailed industries, overseas-linked series, and unresolved remainder within published residual NFB.

Table 3 records how the three mappings treat construction. The Werner adaptation uses the broader BOJ finance-and-insurance series and is labelled a proxy; the Müller–Verner measure is also a BOJ sector adaptation. All three use the same included population, so their differences reflect

Table 3: Literature-anchored credit-taxonomy placements.

Taxonomy	Role	Population	Construction placement	Selection rule
Bezemer–Samarina–Zhang (2020)	primary	Total outstanding loans of domestically licensed banks less local-government loans, exactly exhausted by non-financial business, financial business, property/mortgage, and household non-housing. Domestically licensed describes the lenders, not borrower geography; the residual NFB series includes the explicit overseas-linked component.	Construction is an NFB detail in the primary crosswalk; it is not a stand-alone purpose bucket.	Reported jointly; not selected on outcome fit.
Werner (1997), BOJ proxy	robustness	Same included population as the primary mapping.	Construction enters the Werner-inspired financial-circulation proxy; the available BOJ fit. finance/insurance series is broader than Werner’s original non-bank-financial-institution component.	Reported jointly; not selected on outcome fit.
Muller–Verner (2024), BOJ adaptation	robustness	Same included population as the primary mapping.	Construction is nontradable, following the published tradable/nontradable sector crosswalk.	Reported jointly; not selected on outcome fit.
Legacy G/B/E grouping	archive	Mapped legacy G/B/E borrower groups.	Construction is separate in this legacy grouping.	Archive compatibility only.

The mappings are defined in replication metadata and reported jointly. This is not an externally time-stamped preregistration; the prospective archive applies only to future releases.

classification rather than loan coverage.

3.3 Pre-outcome measurement audits

Aggregation order and granularity. Signed changes are summed inside each fixed bucket before the positive part is taken. Refinements within a bucket leave the result unchanged; moves across buckets do not.

Positive and signed changes. The positive vector describes expansion. The signed vector retains contractions and reconciles to the change in the included stock. Its ratios may fall outside $[0, 1]$, so they are not reported as composition shares.

Purpose and borrower-side comparators. The BOJ consumer-purpose household stock is compared with $L_t^{HHN} = B_t^{HH} - B_t^{HOU}$. It does not define the residual and is not an input to q_t . As a same-provider, narrower measure, it is only a consistency check. MOF and MLIT data provide independent checks for selected branches, not for the full four-bucket composition.

4 Empirical Application

4.1 Same-population reconciliation and coverage

The primary and alternative mappings all exhaust the included stock. Their signed changes also exhaust the change in that stock.

The included stock covers a median 94.5% of the official total; the explicit local-government exclusion accounts for 5.5%. Within residual NFB, detailed industries account for 94.3% at the

Table 4: Published-taxonomy BOJ bridge audits.

Audit		Current readout	Interpretation
Primary population identity	same-	max $ C_t\text{-sum four buckets} =0.000000$; max $ included\ stock\text{-sum four stocks} =0.000000$; max $ signed\text{-scale gap} =0.000000$	Allocation and scale are constructed from one included population. This is an accounting identity, not purpose validation.
Published residual NFB decomposition		median mapped-NFB/NFB=94.3%; median overseas/NFB=5.7%; median unre-solved/NFB=0.001%; max decomposition gap=0.000000	The primary NFB series follows the published Japan residual exactly. Overseas-linked and unresolved components absorbed by that residual are visible rather than silently treated as mapped domestic industries.
Household non-housing split		negative residual rows=0; max split-identity gap=0.000000; consumer-purpose pair N=67, corr=0.904, median absolute stock gap=35947 (JPY 100m)	Household non-housing is total household credit less housing credit. The narrower consumer-purpose series is a same-provider comparator, not independent external validation.
Explicit population coverage	official-	median included/official stock=94.5%; median local-government/official=5.5%; max explicit-scope gap=0.000000	The primary population plus the explicit local-government exclusion reconciles to the official total; coverage does not validate loan purpose.
Literature-anchored taxonomy population identity		Werner max population gap=0.000000; Muller-Verner max population gap=0.000000	The Werner-inspired and Muller-Verner BOJ adaptations are defined in metadata and exhaust the same included population. They are reported jointly; this is not a claim of externally time-stamped preregistration.
Primary vector availability	four-quarter	valid $q_t=64/67$; first=2010-06-30; max $ vector\ sum-1 =0.000000$; max $ q_t\text{-NFB alias gap} =0.000000$	q_t is only the displayed NFB coordinate of the four-part, common-denominator borrower-composition vector.

median, the overseas-linked component for 5.7%, and the unresolved component for 0.001%. Both accounting identities have zero maximum gap at the reported precision.

4.1.1 Positive and net changes

The positive coordinates sum to one, while signed changes reconcile to the included stock change. The four-quarter vector is available for 64 of 67 valid within-taxonomy changes, beginning in 2010Q2. The other endpoints are lost to the rolling window and source availability.

4.1.2 Literature-anchored mappings and limited validation

The household non-housing residual is non-negative in every reported row and its stock identity has zero maximum gap. Its correlation with the narrower BOJ consumer-purpose stock is 0.904 over 67 paired observations, with a median absolute stock gap of 35,947 in units of JPY 100 million. As both series come from the BOJ and cover different concepts, this is a consistency check only. Table 6 adds selected independent checks.

4.2 Use example: pseudo-OOS loss comparison

For horizon $h \in \{4, 8\}$, the use example augments the matched BOJ credit- stock-growth baseline with a reported taxonomy coordinate $z_t^{(r)}$:

$$y_{t,t+h} = \alpha_t + \beta_t g_{t,h}^P + \gamma_t z_t^{(r)} + \varepsilon_{t+h}, \quad (9)$$

where r denotes the Bezemer–Samarina–Zhang NFB coordinate, the Werner financial-circulation coordinate, or the Müller–Verner non-tradable coordinate. Thus Table 5 uses the bridge without treating the three classifications as interchangeable.

The design uses current-vintage data and fixed release lags; historical vintages are not reconstructed. Unavailable forward outcomes are purged, and predictors are standardised within each expanding training window. The moving-block bootstrap uses a block length equal to the horizon and 2,000 fixed-seed replications. Its interval covers the mean difference in standardised squared-error loss, not the difference in RMSE.

Five of the six main-table loss differentials are positive. The four-quarter primary estimate is negative, but its interval includes zero. The Werner four-quarter interval also includes zero, whereas the Müller–Verner interval is above zero; all three eight-quarter intervals are above zero. The present sample therefore gives no reliable evidence that composition improves the forecast.

4.3 Partial external validation

The MOF comparison uses four-quarter changes in manufacturing corporations’ short- and long-term bank borrowing and BOJ manufacturing loan stocks over 2010Q2–2026Q1. Pearson correlation is 0.735, Spearman correlation is 0.792, and change directions agree in 70.3% of 64 common observations. The MLIT FY2024 survey classifies individual housing-loan originations: new and existing acquisition account for 93.5%, refinancing 6.4%, and apartment-loan originations total JPY 3.82 trillion. These sources support selected branches, but their populations do not match the full BOJ measure (Ministry of Finance, 2026; Ministry of Land, Infrastructure, Transport and Tourism, 2026).

Table 5: Bridge application: JP borrower-composition pseudo-OOS loss comparison for the long-term JGB yield change.

h	Literature-anchored coordinate	N	Coordinate-aug. RMSE	Matched-stock RMSE	Mean Δ squared loss [95% CI]
4Q	Bezemer: NFB (primary)	27	1.789	1.847	-0.209 [-1.256, 0.731]
4Q	Werner-inspired BOJ proxy	27	1.921	1.847	0.279 [-0.320, 0.961]
4Q	Müller–Verner: non-tradable (BOJ adaptation)	27	2.045	1.847	0.772 [0.218, 1.332]
8Q	Bezemer: NFB (primary)	15	3.196	2.843	2.130 [0.917, 3.424]
8Q	Werner-inspired BOJ proxy	15	3.366	2.843	3.248 [0.701, 5.661]
8Q	Müller–Verner: non-tradable (BOJ adaptation)	15	3.459	2.843	3.880 [2.109, 5.652]

Current-vintage BOJ inputs use an assumed 90-day lag, and the 2009Q2 classification break is omitted. The matched-stock baseline and each four-quarter coordinate use the same loan population. Forecasts require at least 28 training cases. The moving-block length equals the horizon, with 2,000 fixed-seed replications. The loss differential is candidate standardized squared loss minus baseline loss; negative values favor the candidate. It is not an RMSE difference. Intervals are descriptive and conditional on the displayed specification. Reported forecast origins range from 2018-06-30 to 2024-12-31.

All displayed coordinates are available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). Only augmented models are shown; other specifications are in the companion CSV. No p-values or multiplicity-adjusted results are reported.

For the primary Bezemer four-quarter row, the 4Q mean loss differential is -0.168 with a 20-case minimum and -0.195 with a 24-case minimum; the sign stays negative across all three windows, but this was a selected sensitivity check.

This is an application of the borrower-composition bridge, not a validated forecasting model.

Table 6: Partial external validation of published-bucket branches.

External source	Primary branch	Estimand	Period and readout	Claim boundary
MOF survey	corporate NFB: manufacturing	Correlation of debtor-side and BOJ lender-side 4Q changes in manufacturing borrowing stocks.	2010Q2-2026Q1; Pearson=0.74; man=0.79; tion=70.3%	N=64; Borrower-side convergent evidence only; populations and lender coverage differ, and no loan purpose is observed.
MLIT housing-loan survey	private PROP: household housing	Published purpose composition of individual housing-loan originations and apartment-loan amounts.	FY2024; acquisition=93.5%; refinancing=6.4%; originations=JPY 3.82tn	Direct purpose only for the annual housing branch; it does not validate corporate real-estate loans or quarterly BOJ stock changes.

MOF uses four-quarter differences to reduce sensitivity to its annual April–June sample replacement; this does not make the two credit populations identical. MLIT is annual origination evidence and is therefore not correlated with the quarterly stock-change bridge.

4.4 Implications for data design

A purpose measure would need gross originations classified by both borrower and use, with repayments, write-offs, and refinancing reported separately. Stable taxonomies, data vintages, and reconciliation to a published aggregate would also be needed. Until then, borrower-composition releases should retain their source mapping, positive and net versions, residual accounting, and alternative classifications.

5 Limits

The measure has clear boundaries:

- It observes the sector of the borrower, not the purpose, destination, recipient, collateral, or final expenditure financed by a loan.
- Positive stock changes are not gross originations. Repayment, refinancing, write-offs, and reclassification can change the stocks.
- Granularity invariance holds for refinements within a fixed bucket, not for moving a sector between reported mappings; construction makes this dependence visible.
- Residual NFB includes an overseas-linked component and a small unresolved component. The external sources cover only selected branches under different populations.
- Fixed release lags do not remove look-ahead from historical revisions, and the application sample is too short for predictive or causal inference.

6 Conclusions

The four-bucket crosswalk turns Japanese sectoral loan stocks into a reproducible borrower-composition measure matched to credit scale. Published alternatives show how the result changes when construction is classified differently, while the accounting and coverage checks make the mapping auditable.

The external checks remain partial, and the forecasting application shows no reliable improvement over the matched baseline. The result is a measurement bridge, not a measure of loan purpose or a forecasting result.

Declarations

Data availability. The study uses the public sources listed in Appendix A. An anonymised replication archive accompanies the review files; code and source mappings are linked in the non-anonymous public record.

Conflicts of interest. The authors declare no conflicts of interest.

Declaration of generative AI use. OpenAI Codex assisted with language editing and manuscript organisation. The authors reviewed the output and are responsible for the article.

Appendix A: Reproducibility and Data Sources

Reproducibility. The anonymised review archive regenerates the reported outputs from processed inputs and checks them against expected snapshots. It does not refetch the BOJ source data.

Detailed mapping. The archive supplies the code-level map, series labels, timing information, and alternative assignments. Table 7 records the detailed crosswalk and its residuals.

Table 7: Published-taxonomy BOJ crosswalk and explicit residuals.

BOJ input / source	Bridge role	Construction	Remaining measurement risk	Flow share
Published Japan residual NFB	P_t^{NFB}	Official total less finance/insurance, real estate, local government, and total household loans; construction is included.	The residual also absorbs the separately disclosed overseas-linked and unresolved components.	40.1%
Finance and insurance	P_t^{FIN}	Direct BOJ borrower-sector match to financial business.	Borrower sector does not identify the use of funds.	17.9%
Real estate plus household housing	P_t^{PROP}	Published Japan property/mortgage crosswalk.	Combines a borrower-sector stock with a household-purpose stock.	40.0%
Household total less household housing	P_t^{HHN}	Published residual construction for household non-housing credit.	Includes tax and other household borrowing; it is not labelled consumer credit.	2.1%
Local governments	excluded, explicit	Removed by the published Japan residual formula.	Exclusion changes the population relative to the official total.	6.0%
Overseas-linked loans	NFB component, explicit	Disclosed because the current BOJ total makes it part of residual NFB.	Not a domestic borrower-sector observation.	5.9%
Unresolved residual	NFB component, explicit	Residual after mapped NFB industries and overseas-linked loans.	Has no independent borrower or purpose label.	0.0%
BOJ consumer-purpose loans	validation comparator	Checks, but does not define, the household non-housing residual.	Same provider and narrower concept; not independent external validation.	validation audit
Bezemer, Samarina, and Zhang (2020), Journal of Banking & Finance 113, 105760; Japan crosswalk in DNB Working Paper 559.	literature source	Taxonomy definitions are literature-anchored and all three are reported jointly.	A published crosswalk does not eliminate BOJ measurement error.	not selected on fit

The four primary flow coordinates share one included population. Local government is outside that population. Overseas-linked and unresolved amounts are disclosed as components absorbed by the published NFB residual, not silently discarded.

Appendix B: Legacy Author-Defined Grouping

The earlier G/B/E specification is retained only for reproducibility and archive compatibility. It is not the primary classification and is not used to select the literature-anchored mappings reported in the main text. The full code-level map remains in the replication archive.

Table 8: Legacy G/B/E source map.

BOJ series / code	Borrower	Legacy group	Main limit	Flow share
Outstanding loans by sector, 19 author-defined Group G series: 19 codes in replication metadata; examples: DLLILKG21 / DLLI5DS2TMK; DLLILKG86 / DLLI5DS2TAF; DLLILKG61 / DLLI5DS2TRO	Manufacturing, trade, services, utilities, transport, local governments, and other organisations	Group G: broad sectors, local governments, and other organisations	Borrower sector does not reveal whether funds finance operations, fixed assets, refinancing, or financial assets.	34.4%
Construction outstanding: DLLILKG26 / DLLI5DS2TCT	Construction	Group B: construction	Building finance, land purchase, working capital, and refinancing are not separated.	5.6%
Finance, real-estate, and household outstanding sectors: DLLILKG49 / DLLI5DS2TFI; DLLILKG50 / DLLI5DS2TFX; DLLILKG62 / DLLI5DS2TPN	Finance and insurance; real estate; households	Group E: finance, real estate, and households	Real-estate, finance, housing, and household-consumption uses are mixed.	54.2%
Unclassified outstanding sector: DLLILKG63 / DLLI5DS2TFL	Overseas yen loans and domestic loans transferred overseas	Excluded from domestic borrower composition	Could contain many borrower sectors and loan uses outside the domestic map.	5.8%
New loans for fixed investment by sector: 24 codes in replication metadata; examples: DLLILKG21 / DLLI5DS5TMK; DLLILKG86 / DLLI5DS5TAF; DLLILKG63 / DLLI5DS5TFL	Same sector map as the outstanding-loan series	Within-BOJ purpose-coded comparator	Excludes working capital, refinancing, rollovers, and many asset purchases.	audit only

Positive parts follow within-group aggregation. Joint observations begin in 2009Q2 and the first valid change is 2009Q3. Shares use the selected-series denominator, not the official BOJ total.

Appendix C: Asset-Boom Hypothesis and BOJ Balance-Sheet Acceleration

Exploratory asset-boom hypothesis. A separate literature links broad credit and leverage expansions to asset-price cycles and financial instability (Borio & Lowe, 2002; Jordà et al., 2015; Schularick & Taylor, 2012). The borrower groups used here do not identify asset-directed lending, so the BOJ balance-sheet outcome is retained only as an earlier use case. Only the Müller–Verner eight-quarter estimate is negative, and its interval includes zero. Every other estimate is positive.

Table 9: Auxiliary BOJ balance-sheet acceleration pseudo-OOS check.

h	Literature-anchored coordinate	N	Coordinate-aug. RMSE	Matched-stock RMSE	Mean Δ squared loss [95% CI]
4Q	Bezemer: NFB (primary)	27	1.263	0.853	0.868 [-0.098, 1.913]
4Q	Werner-inspired BOJ proxy	27	1.284	0.853	0.921 [-0.291, 2.659]
4Q	Müller–Verner: non-tradable (BOJ adaptation)	27	0.942	0.853	0.160 [-0.100, 0.501]
8Q	Bezemer: NFB (primary)	15	0.534	0.360	0.156 [0.023, 0.291]
8Q	Werner-inspired BOJ proxy	15	0.803	0.360	0.516 [0.258, 0.774]
8Q	Müller–Verner: non-tradable (BOJ adaptation)	15	0.318	0.360	-0.029 [-0.060, 0.003]

Current-vintage BOJ inputs use an assumed 90-day lag, and the 2009Q2 classification break is omitted. The matched-stock baseline and each four-quarter coordinate use the same loan population. Forecasts require at least 28 training cases. The moving-block length equals the horizon, with 2,000 fixed-seed replications. The loss differential is candidate standardized squared loss minus baseline loss; negative values favor the candidate. It is not an RMSE difference. Intervals are descriptive and conditional on the displayed specification. Reported forecast origins range from 2018-06-30 to 2024-12-31.

All displayed coordinates are available in 64/67 post-break source quarters (missing-source pattern Q1:1,Q3:1,Q4:1). Only augmented models are shown; other specifications are in the companion CSV. No p-values or multiplicity-adjusted results are reported.

This appendix records the asset-acceleration check separately from the main application.

References

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